

SIH26157: SAT-SA Project Report

Yeh report simple words mein explain karti hai ki NCIIPC ka problem statement asal mein kya maang raha hai, aur hamare banaye hue prototype ne unki demands ko kaise satisfy kiya hai.

1. Problem Statement Kya Maang Raha Hai? (The Demand)

NCIIPC (National Critical Information Infrastructure Protection Centre) ka kaam hai desh ke bade sectors (jaise Power, Banking, Telecom) ki cyber security (SOCs) ko monitor karna.

Main Problem: NCIIPC ke supervisors manually check karte hain ki kya ye SOCs sach mein theek se kaam kar rahe hain ya nahi. Bahut baar SOCs ke dashboard par sab kuch 'Green' aur safe dikhta hai, par andar ki reality kuch aur hoti hai (jaise alerts ko bina check kiye close kar dena). Is manual checking mein bahut time lagta hai.

NCIIPC ko ek Supervisory Analytics Tool (SAT-SA) chahiye jo:

1. Execution Gaps pakde: Matlab rules toh bane hain, par log unko follow nahi kar rahe. (E.g., Alert aaya aur bina investigate kiye turant close kar diya).
2. Negative Space pakde: Matlab jo cheez wahan honi chahiye thi, par missing hai. (E.g., Kisi critical server se koi data ya log aa hi nahi raha).
3. Fully Offline ho: Tool ko internet ya cloud ki zaroorat nahi honi chahiye (security reasons ke liye).
4. Explainable ho: AI ya system jo bhi anomaly pakde, wo supervisor ko clear reason bataye ki 'kyun' usne isko flag kiya.

2. Humne Demand Ko Kaise Satisfy Kiya? (Our Solution)

Humne ek 2-Layer detection engine aur ek fully offline dashboard banaya hai.

A. Execution Gaps & Negative Space (Simple Examples)

Humare system ne in problems ko solve karne ke liye Rules aur Machine Learning (ML) ka use kiya:

* Demand: 'Critical alerts closed without escalation' (Execution Gap)

* Our Solution (Layer A): Humne ek rule banaya 'critical_no_escalation'. Agar koi SOC kisi 'Critical' severity ke alert ko bina kisi higher authority ko bheje (escalate kiye) close karta hai, toh hamara tool usko turant flag kar dega.

* Demand: 'Repetitive or template-driven investigations' (Execution Gap)

* Our Solution (Layer A): Humne Natural Language Processing (NLP) use kiya. Agar analyst baar-baar ek hi 'Copy-Paste' comment daal kar cases close kar raha hai, toh hamara 'templated_investigation' rule usey pakad lega.

* Demand: 'Peer Benchmarking & Outliers'

* Our Solution (Layer B): Humne Z-Scores aur Isolation Forest (Machine Learning) ka use kiya. Agar Sector 1

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ke saare banks ek alert ko solve karne mein average 2 ghante lagate hain, par ek bank usko 2 minute mein close kar raha hai, toh hamara AI model us abnormal behaviour ko pakad lega.

B. Explainability (Reason Dena)

NCIIPC nahi chahta ki tool sirf 'Risk' bol de. Supervisor ko evidence chahiye.

* Our Solution: Hamare Dashboard mein ek 'Finding Dossier' page hai. Jab bhi tool koi galti pakadta hai, wo ek human-readable Explanation deta hai (e.g., 'This critical alert was closed in 1.5 mins without escalation') aur sath hi original JSON Data (Evidence) bhi screen par dikhata hai.

C. Deployment & Offline Constraint

* Demand: No Internet, No Cloud.

* Our Solution: Humara pura system Python Flask par bana hai jo localhost (aapke apne computer) par chalta hai. UI ko sundar banane ke liye humne Chart.js aur CSS ka use kiya hai jo locally bundled hain. Internet band karne par bhi yeh 100% chalega.

3. Project Dashboard (Screenshots)

(Kyunki ye ek PDF document hai, aap apne project ko run karke unke screenshots yahan attach kar sakte hain presentation mein).

I. Command Center (Overview): Yahan supervisor saare sectors ka risk ranking aur total anomalies dekh sakta hai.

II. Entity Dossier & Alert Trends: Kisi ek company par click karne par uski details aur 90-days ka alert graph dikhta hai.

III. Explainable Evidence (Raw Dossier): Left side mein human-readable explanation hai aur right side mein raw JSON evidence hai jo supervisor audit ke liye export kar sakta hai.